

Problem Set 4: POLS 3316

1. What rule helps us tie the results of a large number of trials to the normal distribution? A
 - a. The Central Limit Theorem
 - b. The 68-95-99.7 rule
 - c. The Pythagorean Theorem
 - d. The Law of Large Numbers
2. What helps us find probabilities in a normal distribution based on standard deviations? C
 - a. The Standard Deviation Rule
 - b. Standard Errors
 - c. The 68-95-99.7 Rules
 - d. The Law of Large Numbers
3. The mean is the sum of the observations divided by the number of observations.
4. The standard deviation is the square root of the variance.
5. The variance is the average squared distance to the mean.
6. Find the correlation of your X and Y project variables. (This was done in Lab 7, including demonstration of two possible errors. If you have missing data points, NA or NaN values, you'll need to include `use = "pairwise.complete.obs"` in the `cor()` command. If you get a non-numeric variable error, you'll need to convert your variable to numeric, as done with the `episodes` variable.) **You can do this in R Studio or your R Studio Cloud project space, just copy and paste the code and result in a text area. Don't copy the code chunk "{r}" of you will get an error.**

ANSWER VARIES - needs code and result

7. Find the sum, mean, and population variance of (5,10,15,20) sum: $5 + 10 + 15 + 20 = 50$ mean: $50/4 = 12.5$ population variance: 31.25

$$\frac{(5-12.5)^2 + (10-12.5)^2 + (15-12.5)^2 + (20-12.5)^2}{4}$$

or

```
vec1 <- c(5,10,15,20)
mean1 <- mean(vec1)
squares1 <- sum((vec1 - mean1)^2)
var1 <- squares1/4
var1
```

```
## [1] 31.25
```

Bonus: Write a loop of your choice (*for loop* or *while loop*) to print the even numbers through ten. (Hint: the easiest way is to print the result of a multiplying something times 2.)

```
for (i in 1:5) {  
  print(i*2)  
}
```

```
## [1] 2  
## [1] 4  
## [1] 6  
## [1] 8  
## [1] 10
```